

# Weather Project: Seattle vs Boston

## 1. Introduction

Seattle is widely known for its rainy weather, but does it actually rain more in Seattle than in Boston? This project uses daily precipitation data from Seattle, WA and Boston, MA from January 1, 2018 to December 31, 2022 to compare precipitation between the two cities. The analysis examines the amount and frequency of precipitation in each city.

## 2. Data

This project uses historical daily precipitation data for Seattle, WA and Boston, MA from January 1, 2018 to December 31, 2022 from National Oceanic and Atmospheric Administration (NOAA) (source: [Climate Data Online Search](#)).

## 3. Analysis Methods

First, I cleaned and combined the daily precipitation data for Seattle and Boston into a clean CSV file. Next, I compared the two cities by looking at the mean precipitation by day and month, the total precipitation by year, the proportion of days with any precipitation, and the mean precipitation on days when precipitation occurred. Finally, I used statistical tests to determine whether the differences in mean precipitation and proportion of days with precipitation between Boston and Seattle are statistically significant each month.

## 4. Results

### 4.1 How does average precipitation compare between Seattle and Boston?

In terms of the mean precipitation by day, Boston has higher precipitation than Seattle (0.15 inches vs 0.11 inches). In terms of the mean daily precipitation by month, Seattle has more rain in November, December, and January than Boston. However, Seattle is drier during summer. Boston's precipitation is more consistent across the year. Therefore, we can conclude that Seattle shows stronger seasonality than Boston.

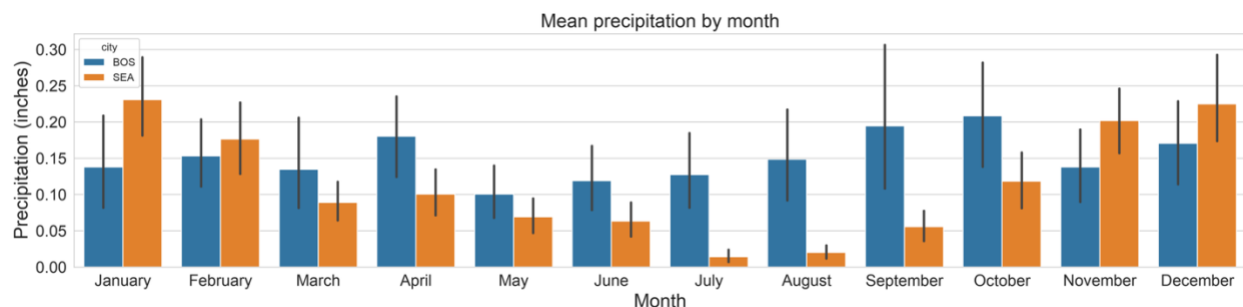


Figure 1. Mean daily precipitation by month

### 4.2 How does total annual precipitation compare between Seattle and Boston from 2018 to 2022?

Boston had almost twice as much total precipitation as Seattle in 2018, while the two cities had very similar totals in 2020. Overall, Boston had higher total annual precipitation than Seattle. Boston also had greater variability in total annual precipitation (43.66-70.5 inches), whereas Seattle's total annual precipitation was relatively stable (37.24-44.43 inches).

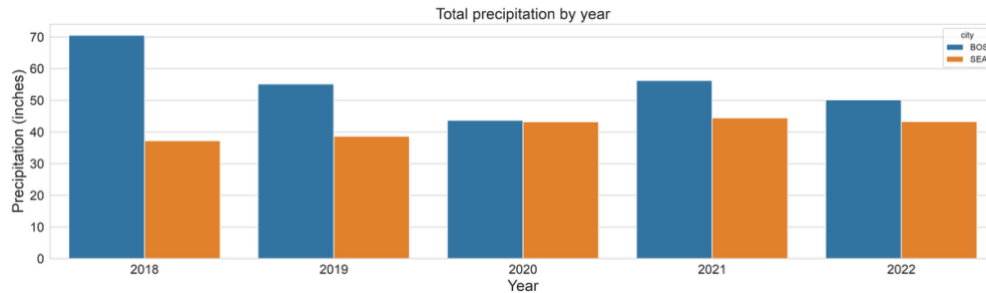


Figure 2. Total precipitation by year

#### 4.3 How often does precipitation occur in Seattle and Boston?

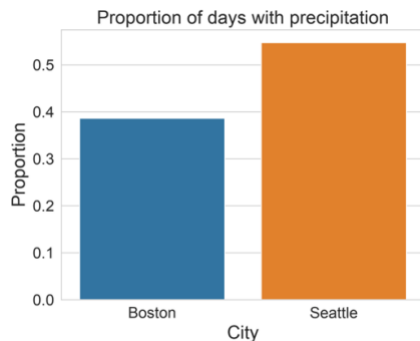


Figure 3. Proportion of days with precipitation

According to this chart, more than half of the days in Seattle had precipitation, while about 40% of the days in Boston had precipitation. Therefore, Seattle has precipitation more frequently than Boston.

#### 4.4 When precipitation occurs, how much does each city receive?

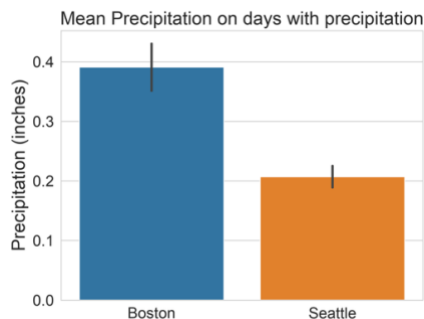


Figure 4. Mean precipitation on days with precipitation

Although Seattle has precipitation more frequently, Boston has more precipitation on days with precipitation. On days with precipitation, Boston averages about 0.39 inches, compared with about 0.21 inches in Seattle.

To evaluate whether these observed differences were statistically meaningful, the tests show that the mean daily precipitation differs significantly between Boston and Seattle during January, April, June, July, August, September, and October. The proportion of days with precipitation differs significantly during January, February, March, July, November, and December.

### 5. Conclusion

Overall, whether it rains more in Seattle than Boston depends on how we define “rain more.” Seattle has precipitation more frequently than Boston. Seattle also shows stronger seasonality, with particularly frequent precipitation during the winter months. However, Boston has more precipitation on days when precipitation occurs. Boston also has a higher average daily precipitation and higher annual precipitation totals. Therefore, in terms of frequency, Seattle can be considered rainier. In terms of the amount of precipitation, Boston can be considered rainier.